

George Vasilakopoulos

AI & Software Engineer

Zurich, Switzerland

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Summary

AI and Software Engineer with 5+ years of experience focused on production-grade AI platforms, LLM systems, and distributed backend infrastructure. Experienced in building and deploying scalable enterprise systems and data pipelines in cloud environments.

Experience

AI & Software Engineer — UBS Zurich

02/2024 – Present

Legal Tech

Document Redaction Platform [PROD]

- Architected a distributed, event-driven document processing platform for automated detection and redaction of sensitive information across large-scale PDF/DOCX workflows using OpenAI LLM
- Designed asynchronous, queue-based processing architecture using FastAPI, Azure Functions, and Azure Storage Queues to decouple processing, redaction, and validation stages
- Built human-in-the-loop review workflows integrated into the processing pipeline for compliance-critical decisioning and auditability in regulated legal environments
- Engineered scalable storage and compute layers using PostgreSQL, Azure Blob Storage, and Terraform with GitLab CI/CD for reliable multi-environment deployments

Enterprise Semantic Search & Retrieval Platform [PROD]

- Designed and implemented a distributed LLM-powered retrieval system (RAG-style architecture) for semantic search across millions of legal documents
- Built end-to-end ingestion and indexing pipelines including document chunking, embedding generation, and hybrid vector + keyword indexing for retrieval
- Developed microservices-based, event-driven architecture using Azure Functions, Service Bus, Cosmos DB, and Azure AI Search for scalable data processing
- Optimized retrieval performance through asynchronous processing, pipeline parallelization, and indexing strategies for production-scale workloads
- Tuned system for low-latency query serving and high-throughput ingestion across enterprise-wide legal knowledge bases

AI & Data Engineer — Credit Suisse Zurich

01/2023 – 01/2024

Central Analytics

Enterprise ML Platform — Distributed MLOps System [PoC]

- Designed a cloud-native distributed ML platform architecture for enterprise data quality evaluation
- Built orchestrated ML workflows using Prefect for scheduled pipeline execution across heterogeneous data sources
- Developed FastAPI-based microservices for model serving and orchestration, enabling standardized deployment interfaces
- Deployed containerized workloads on OpenShift/Kubernetes using Docker with CI/CD automation via Jenkins
- Established modular ML pipeline architecture supporting experimentation, monitoring, and production deployment workflows

Full Stack / Quant Engineer — Credit Suisse Zurich

11/2021 – 12/2022

Private Equity Markets

Private Equity Reporting System [PROD]

- Architected a high-throughput enterprise reporting system generating 400+ automated client reports monthly for private equity portfolios
- Designed backend data pipelines using .NET (C#), MS SQL Server, and SSRS for structured financial reporting workflows

- Built deterministic reporting pipelines ensuring data consistency, auditability, and regulatory compliance for institutional clients
- Owned production reporting infrastructure used by investment teams for mission-critical portfolio reporting and decision support

Portfolio Forecasting Platform [PROD]

- Developed a quantitative portfolio analytics platform for forecasting and scenario analysis across investment portfolios
- Implemented stochastic financial models including Cox–Ingersoll–Ross (CIR) and Yale long-horizon portfolio models
- Built API-driven model serving layer using Flask, integrating quantitative models into Power BI dashboards for analytics
- Designed modular forecasting engine supporting scenario simulation, risk analysis, and forward-looking portfolio evaluation workflows

AI Researcher — IBM Research Zurich

05/2021 – 10/2021

Computational Systems Biology

TCR-Antigen Binding Model

- Researched semi-supervised variational autoencoders for biological sequence modeling
- Applied NLP and deep learning to TCR-antigen interaction prediction
- Developed representation learning methods for computational biology applications

Education

ETH Zurich — MSc Data Science

09/2018 – 11/2021

Grade: 5.55 / 6

Focus Areas: Advanced Machine Learning, Big Data Systems, Data Management Systems, Statistical Modeling, NLP, Computational Intelligence, Machine Learning for Healthcare

University of Patras — BSc Mathematics

09/2013 – 07/2018

Grade: 8.81 / 10 (Top 2% of class)

Focus Areas: Probability Theory, Statistics, Operational Research

Thesis: Game Theoretical Aspects of Queueing Systems

Skills

AI/ML: LLM systems (RAG, embeddings, semantic search), NLP, probabilistic modeling

Backend Engineering: Python, FastAPI, .NET (C#), REST APIs, Microservices

Data Engineering: ETL pipelines, distributed data processing

Cloud & Systems: Azure, distributed systems, event-driven architectures, microservices, async processing, queue-based systems, scalable backend design

DevOps: Kubernetes, Docker, Terraform, CI/CD (GitLab, Jenkins)

Data & Analytics: SQL, NoSQL databases, Power BI, SSRS

Exams & Achievements

GRE Quantitative: 169/170

Mathematics Panhellenic Exam: 93% (Top 2% in Greece)

Additional

Mentorship: Job Coach — Powercoders (supporting young professionals)

Languages: Greek (Native), English (Proficient)